

Bond lengths from spectroscopy, from theory, and from both together

A test on three monofluorinated molecules with published structures

Short version. Quantum chemistry alone puts the carbon–fluorine bond 27 to 31 thousandths of an Ångström too long in every one of the three molecules tested, always in the same direction — a systematic bias, not scatter, and exactly the kind of error measurements should be able to fix. They partly do: on the C–F bond, some combination of data and theory beats theory alone in all 6 cases tested. On overall bond accuracy the result is weaker — the better of the two ways of combining them wins in 5 of 6, even when that better way is picked with hindsight, and no rule was found that picks it in advance.

Two things explain the gap. First, every one of these molecules is underdetermined *even with every published rotational constant included* — vinyl fluoride's 22 measured constants carry only 9 independent constraints on 12 structural parameters. Second, the measurements and the reference structure are different kinds of quantity, and the half-percent mismatch between them is currently uncorrected; that, rather than the choice of algorithm, is what limits these results. An earlier version of this experiment used isotopologue constants derived from the reference structures instead of measured ones, and reached a tidier and more favourable conclusion. It does not survive real data, and section 7.4 says how.

1. What was being tested

A molecule's structure — which atoms are how far apart, and at what angles — can be worked out two quite different ways.

From theory. Solve the equations of quantum mechanics for the molecule's electrons and find the atomic arrangement with the lowest energy. This needs no experiment at all, but the answer is only as good as the approximations used, and cheap approximations have systematic biases.

From experiment. Measure how fast the molecule tumbles in the gas phase. Microwave spectroscopy gives three numbers per molecule — the rotational constants A, B and C — which are fixed by how the mass is distributed. Swap one atom for a heavier isotope and the mass distribution changes in a known way, giving three more numbers. Enough isotopic variants and the structure can be solved for.

The catch with the experimental route is that three numbers per variant is not many. A seven-atom molecule has fifteen independent structural parameters; one measured species gives three numbers. The problem is underdetermined, and there is a whole family of structures that fit the data equally well.

The hybrid idea. Use the data for the directions the data can see, and use theory for the rest. This report asks whether that actually works — specifically, whether the combination beats theory on its own, which is the practical question, because theory on its own is always available and costs nothing but computer time.

Fluorinated molecules make this sharper than usual. Fluorine has only one stable isotope, so no isotopic substitution ever moves the fluorine atom. Its position has to be inferred indirectly, from the requirement that the molecule's centre of mass sits where the measurements say it does, once every other atom has been located. That makes the C–F bond the hardest parameter in the molecule and a good stress test.

2. The four methods compared

Method	What it uses	What it does
theory	no experimental data	Minimises the quantum-chemical energy. The answer depends only on the molecule, not on any measurement.

Method	What it uses	What it does
experiment	rotational constants only	Adjusts the geometry until the calculated rotational constants match the measured ones. Directions the data cannot see are left to drift.
hybrid (split)	both	Splits the parameter space in two using an SVD of the sensitivity matrix. Data owns every direction it can see; theory owns the remainder outright.
hybrid (joint)	both	Leaves every direction contested and weights the two sources by how well each knows that direction. Theory's weight is set by the distance over which it is trusted, here 0.005 Å.

The two hybrids differ only in how the two sources are merged. Both are run for every molecule and every data level, because which one is better turns out not to be predictable in advance (section 7.4).

3. The molecules, and why these three

Each molecule needed a published experimental structure *and* published measured rotational constants for a useful set of its isotopic variants. The two must be mutually consistent: feeding the published geometry back through the rotational-constant calculation should reproduce every measured constant to within about 1%, the size of the unavoidable zero-point-motion offset. A larger gap means the structure and the data disagree, or a variant has been assigned to the wrong atom — and the comparison would then be measuring that error rather than anything about the methods. Every species was checked individually, not just the parent, which is how the mislabelled vinyl fluoride species in section 4 were caught.

Molecule	Formula	Atoms	Structural parameters	Measured species	Measured constants	Worst disagreement
Vinyl fluoride	CH ₂ =CHF	6	12	8	22	1.37%
Acetyl fluoride	CH ₃ COF	7	15	10	30	0.32%
Fluoroethane	CH ₃ CH ₂ F	8	18	6	18	0.69%

The last column is the worst disagreement between the published geometry and any one of its measured constants. Sources — **Vinyl fluoride**: structure, Hayashi & Inagusa, J. Mol. Spectrosc. 138, 135 (1989); CCCBDB; constants, NBS Monograph 70 Vol. IV p.183 (entry 780); **Acetyl fluoride**: structure, Pierce & Krisher, J. Chem. Phys. 31, 875 (1959); CCCBDB; constants, NBS Monograph 70 Vol. III p.119 (entry 390); **Fluoroethane**: structure, Kraitichman & Dailey (1955); CCCBDB; constants, NBS Monograph 70 Vol. III p.158 (entry 550).

A fourth candidate, formyl fluoride (HCOF), was checked and dropped. Its tabulated geometry and its tabulated constants disagree by 6.2% — far beyond any zero-point effect — so at least one of them is wrong and it cannot serve as a ground truth. It is mentioned here because the selection test is the part of the setup most likely to be skipped, and skipping it would have silently poisoned a quarter of the results.

4. How the test was run

Level of theory

Hartree–Fock with the 6-31G basis set, through PySCF, for every molecule and every method — including the quantum half of both hybrids. This is deliberately a modest level of theory. The point is not to produce the best possible structures; it is to compare three ways of using the same quantum surface, and holding the surface fixed is what makes the comparison mean anything. It is also the same level used on fluorobenzene earlier in this project, so those numbers are comparable with these.

Starting geometry

Every method starts from the same deliberately wrong geometry: the published structure with random noise of 0.03 Å added to every coordinate (one fixed random seed, so the runs are reproducible). Without this, a method could look good simply by not moving. The starting error is reported in every table as the do-nothing baseline that any method has to beat.

The rotational constants

Every constant used here is a measured literature value, transcribed from *NBS Monograph 70, Microwave Spectral Tables* (National Bureau of Standards, 1968–69) — the standard compilation of the microwave work of that era. Nothing is back-calculated from a geometry. Where the original study did not determine a constant it is left out rather than filled in: two of the eight vinyl fluoride species have no measured A at all, and several others have A quoted to three or four significant figures against seven for B and C. That unevenness is real, and the fit is told about it through per-constant uncertainties rather than having it smoothed away.

The uncertainties are set by model error, not measurement error. The reference structures are substitution (r_s) structures while the constants are ground-state (r_0) values, and that difference is worth a few tenths of a percent — far more than the 0.01 MHz to which B and C are quoted. A is given twice the floor of B and C, because it is both the least well determined constant of a near-prolate top and the one most sensitive to zero-point out-of-plane motion; the measured residuals show exactly that pattern.

One correction to the source. The compilation's cis/trans labels on the two doubly-deuterated vinyl fluoride species (ids 789 and 791) contradict its own constants. Deuterating the CHF hydrogen shifts B by only about 1 MHz, so B alone identifies which CH_2 hydrogen carries the second deuterium — and on that test the labels are swapped. Assigned by the constants, both species match the published geometry to better than 0.4%; assigned by the labels they are off by 6 to 22%, which is impossible for an isotopic substitution. The assignments here follow the constants. Every species is re-checked this way by `scripts/check_monofluoro_references.py` before any fitting is done.

Two data levels, and how much they actually constrain

Counting measurements overstates how much is known. Three constants from one species are not three independent constraints on the structure: for a planar molecule the third is nearly fixed by the first two, since the inertial defect is close to zero. The honest measure is the rank of the stacked sensitivity matrix — how many independent directions in the structure the data can actually see. It is computable in advance, from the starting geometry, without knowing the answer.

Data level	Species	Constants measured	Independent constraints	Parameters	Unconstrained
Vinyl fluoride — parent only	1	3	2	12	10
Vinyl fluoride — all species	8	22	9	12	3
Acetyl fluoride — parent only	1	3	3	15	12
Acetyl fluoride — all species	10	30	14	15	1
Fluoroethane — parent only	1	3	3	18	15
Fluoroethane — all species	6	18	15	18	3

Vinyl fluoride is the clearest case: 22 measured constants carry only 9 independent constraints on 12 parameters. **Every molecule here is underdetermined even with every published constant included** — which is the situation the hybrid exists for, and not the comfortable regime raw counting would suggest.

How error is measured

Each fitted geometry is reduced to chemically meaningful numbers — bond lengths and bond angles, with symmetry-equivalent ones averaged — and compared with the same numbers from the published structure. Bond errors are quoted in milli-Ångström ($\text{m}\text{\AA}$, thousandths of an Ångström); for scale, a carbon–carbon bond is about 1500 $\text{m}\text{\AA}$ long and a good structure determination is accurate to a few $\text{m}\text{\AA}$. Angle errors are in degrees. The C–F bond error is reported separately and with its sign, because it is the parameter the whole exercise is about.

5. Results

Three tables summarise all 21 fits. The first is the headline number — overall bond-length accuracy. The second isolates the carbon–fluorine bond. The third covers angles.

5.1 Overall bond-length error (root-mean-square)

Molecule	Data level	start	theory	experiment	hybrid (split)	hybrid (joint)
Vinyl fluoride	parent only	22.1	19.0	21.0	20.3	19.5
	all species	22.1	19.0	31.1	31.1	18.6
Acetyl fluoride	parent only	35.5	15.8	34.5	32.1	15.5
	all species	35.5	15.8	8.8	8.8	15.1
Fluoroethane	parent only	22.8	15.0	14.2	8.5	14.6
	all species	22.8	15.0	6.9	6.6	14.1

Lower is better; the *start* column is the deliberately wrong geometry every method began from, so any method above its own start column has made things worse. All values in mÅ; the best of the four methods in each row is highlighted.

5.2 The carbon–fluorine bond, signed

Molecule	Data level	start	theory	experiment	hybrid (split)	hybrid (joint)
Vinyl fluoride	parent only	+10.1	+30.7	-2.4	-3.5	+29.2
	all species	+10.1	+30.7	-10.3	-10.3	+29.1
Acetyl fluoride	parent only	+51.4	+28.8	+36.2	+34.3	+27.8
	all species	+51.4	+28.8	+11.7	+11.7	+25.1
Fluoroethane	parent only	+7.3	+26.9	-0.8	+9.4	+25.1
	all species	+7.3	+26.9	+9.0	+8.6	+22.2

A positive number means the bond came out too long. Theory is positive and large in all three molecules. All values in mÅ; the best of the four methods in each row is highlighted.

5.3 Bond-angle error (root-mean-square)

Molecule	Data level	start	theory	experiment	hybrid (split)	hybrid (joint)
Vinyl fluoride	parent only	3.37	1.60	1.94	1.81	1.47
	all species	3.37	1.60	6.28	6.28	1.51
Acetyl fluoride	parent only	0.72	1.19	1.19	1.17	1.16
	all species	0.72	1.19	0.92	0.92	1.28
Fluoroethane	parent only	2.04	0.63	0.89	0.38	0.68
	all species	2.04	0.63	0.83	0.84	0.66

Angles behave far less consistently than bond lengths: no method wins across the board, and theory is competitive here in a way it never is for the C–F bond. All values in degrees; the best of the four methods in each row is highlighted.

6. Molecule by molecule

The tables below give every bond length and bond angle, for the published structure and for each method, at both data levels. Symmetry-equivalent parameters are averaged into one row.

Vinyl fluoride — CH₂=CHF

6 atoms, 12 structural parameters. Published B₀ = 64,582.7, 10,636.8, 9,118.2 MHz; inertial defect +0.09 amu Å² (planar, as expected). The published geometry reproduces its own constants to 1.37%.

parent only (3 numbers)	published	start	theory	experiment	hybrid (split)	hybrid (joint)
C1=C2	1.3290	1.3387	1.3102	1.3213	1.3227	1.3079
C1-F	1.3470	1.3571	1.3777	1.3446	1.3435	1.3762
C1-H4	1.0820	1.1023	1.0673	1.1231	1.1216	1.0669
C2-H5	1.0870	1.0842	1.0708	1.0879	1.0879	1.0705
C2-H6	1.0770	1.1198	1.0704	1.0984	1.0978	1.0663
F-C1=C2	120.80	116.66	121.22	121.88	121.88	120.97
H4-C1=C2	129.20	134.02	127.39	131.71	131.47	127.60
F-C1-H4	110.00	109.04	111.39	106.41	106.65	111.43
C1=C2-H5	119.00	123.08	121.51	120.04	120.01	121.27
C1=C2-H6	120.90	117.98	120.19	120.94	120.90	120.31
H5-C2-H6	120.10	118.85	118.30	119.02	119.09	118.42

Bond lengths in Ångström, angles in degrees (shaded rows). Closest to published is highlighted.

all species (22 numbers)	published	start	theory	experiment	hybrid (split)	hybrid (joint)
C1=C2	1.3290	1.3387	1.3102	1.3389	1.3389	1.3095
C1-F	1.3470	1.3571	1.3777	1.3368	1.3368	1.3761
C1-H4	1.0820	1.1023	1.0673	1.0874	1.0874	1.0679
C2-H5	1.0870	1.0842	1.0708	1.0491	1.0491	1.0710
C2-H6	1.0770	1.1198	1.0704	1.1332	1.1332	1.0702
F-C1=C2	120.80	116.66	121.22	121.44	121.44	120.45
H4-C1=C2	129.20	134.02	127.39	120.15	120.15	127.78
F-C1-H4	110.00	109.04	111.39	118.40	118.41	111.78
C1=C2-H5	119.00	123.08	121.51	121.64	121.64	121.29
C1=C2-H6	120.90	117.98	120.19	113.53	113.53	120.28
H5-C2-H6	120.10	118.85	118.30	124.83	124.83	118.43

Bond lengths in Ångström, angles in degrees (shaded rows). Closest to published is highlighted.

Acetyl fluoride — CH₃COF

7 atoms, 15 structural parameters. Published B₀ = 11,039.3, 9,685.6, 5,322.1 MHz; inertial defect -3.00 amu Å² (non-planar, as expected for a methyl group). The published geometry reproduces its own constants to 0.32%.

parent only (3 numbers)	published	start	theory	experiment	hybrid (split)	hybrid (joint)
C1-C2	1.5030	1.5313	1.4852	1.5114	1.5078	1.4838
C1=O	1.1810	1.1425	1.1876	1.1370	1.1435	1.1856
C1-F	1.3480	1.3994	1.3768	1.3842	1.3823	1.3758
C2-H5	1.0840	1.1192	1.0771	1.1299	1.1295	1.0773
C2-H6	1.0840	1.0715	1.0821	1.0610	1.0621	1.0826
O=C1-F	121.35	121.43	119.34	122.31	121.64	119.40
O=C1-C2	128.35	128.79	129.67	128.76	129.64	129.58
F-C1-C2	110.30	109.59	111.00	108.83	108.72	111.02
C1-C2-H5	110.40	109.11	109.83	109.63	110.04	109.76
C1-C2-H6	110.40	110.85	109.73	112.21	111.98	109.71

Bond lengths in Ångström, angles in degrees (shaded rows). Closest to published is highlighted.

all species (30 numbers)	published	start	theory	experiment	hybrid (split)	hybrid (joint)
C1-C2	1.5030	1.5313	1.4852	1.4929	1.4929	1.4825
C1=O	1.1810	1.1425	1.1876	1.1875	1.1875	1.1868
C1-F	1.3480	1.3994	1.3768	1.3597	1.3597	1.3731
C2-H5	1.0840	1.1192	1.0771	1.0749	1.0749	1.0767
C2-H6	1.0840	1.0715	1.0821	1.0889	1.0889	1.0820
O=C1-F	121.35	121.43	119.34	119.96	119.96	119.20
O=C1-C2	128.35	128.79	129.67	129.13	129.13	129.90
F-C1-C2	110.30	109.59	111.00	110.92	110.92	110.91
C1-C2-H5	110.40	109.11	109.83	110.23	110.23	109.88
C1-C2-H6	110.40	110.85	109.73	109.26	109.26	109.69

Bond lengths in Ångström, angles in degrees (shaded rows). Closest to published is highlighted.

Fluoroethane — CH₃CH₂F

8 atoms, 18 structural parameters. Published B_0 = 36,070.3, 9,364.5, 8,199.7 MHz; inertial defect -6.34 amu Å² (non-planar, as expected for a methyl group). The published geometry reproduces its own constants to 0.69%.

parent only (3 numbers)	published	start	theory	experiment	hybrid (split)	hybrid (joint)
C1-C2	1.5050	1.5273	1.5091	1.5207	1.5088	1.5068
C1-F	1.3980	1.4053	1.4249	1.3972	1.4074	1.4231
C1-H4	1.0950	1.0893	1.0786	1.0990	1.0960	1.0780
C2-H6	1.0910	1.1330	1.0839	1.1146	1.1024	1.0822
C2-H7	1.0900	1.0738	1.0820	1.0763	1.0788	1.0821
F-C1-C2	109.70	106.03	108.83	109.33	109.38	108.77
F-C1-H4	106.04	105.66	106.62	105.18	106.25	106.61
C2-C1-H4	112.90	113.97	112.42	113.77	112.80	112.39
H4-C1-H5	108.81	110.64	109.60	108.82	108.95	109.73
C1-C2-H6	109.70	107.09	110.34	108.56	109.46	110.39
C1-C2-H7	110.40	110.24	110.58	109.05	109.59	110.51

Bond lengths in Ångström, angles in degrees (shaded rows). Closest to published is highlighted.

all species (18 numbers)	published	start	theory	experiment	hybrid (split)	hybrid (joint)
C1-C2	1.5050	1.5273	1.5091	1.5026	1.5028	1.5053
C1-F	1.3980	1.4053	1.4249	1.4070	1.4066	1.4202
C1-H4	1.0950	1.0893	1.0786	1.1033	1.1031	1.0774
C2-H6	1.0910	1.1330	1.0839	1.0996	1.0994	1.0804
C2-H7	1.0900	1.0738	1.0820	1.0872	1.0874	1.0810
F-C1-C2	109.70	106.03	108.83	109.80	109.79	108.88
F-C1-H4	106.04	105.66	106.62	106.07	106.07	106.63
C2-C1-H4	112.90	113.97	112.42	113.55	113.58	112.29
H4-C1-H5	108.81	110.64	109.60	107.33	107.25	109.80
C1-C2-H6	109.70	107.09	110.34	108.71	108.74	110.14
C1-C2-H7	110.40	110.24	110.58	109.68	109.74	110.58

Bond lengths in Ångström, angles in degrees (shaded rows). Closest to published is highlighted.

7. What the numbers show

7.1 Theory has a consistent, one-directional C–F bias

Across the three molecules, theory puts the C–F bond +31, +29, +27 mÅ from the published value. Same sign, similar size, every time: at this level of theory the C–F bond is systematically too long by roughly 30 mÅ. That is not random error, it is a bias of the method, and no amount of tightening the geometry optimisation will remove it. It is exactly the kind of error experimental data is well placed to correct, and it is the single clearest argument for combining the two sources at all.

7.2 More data does not always help

The most surprising result here is vinyl fluoride. Fitting the parent's three constants gives a bond error of 21 mÅ. Adding every other measured species — seven more, twenty-two constants in total — makes it *worse*, 31 mÅ, and worse than the deliberately wrong geometry the fit started from. Some angles move by nine degrees.

This is not a bug, and checking that mattered. The published structure reproduces its own measured constants with a reduced chi-squared near one, so the data and the reference agree; the fit simply finds a structure that fits the numbers *better* than the true one does. It can, because 22 constants carry only 9 independent constraints on 12 parameters, and because the residuals are systematically one-signed — B and C sit about 0.5% off in the same direction for every species, which is the r_s -versus- r_0 offset described in section 4. The fit spends its three unconstrained directions removing that bias, and pays for it in structural distortion. More data made the pull stronger without closing the directions it could pull through.

7.3 Spectroscopy alone is unreliable on thin data

With one measured species — three numbers against twelve to eighteen parameters — fitting to the data gives bond errors of 21, 34, 14 mÅ, starting from 22, 36, 23 mÅ. In two of the three the fit has barely improved on the deliberately wrong geometry it began with, moving it about 1 mÅ.

This is not the optimiser failing; it is the underdetermination showing through. Much of the damage lands on bonds involving hydrogen — the worst is off by +41, +46, +24 mÅ — because moving a hydrogen barely shifts the moments of inertia, so the fit will move one a long way to buy a small improvement. But it is not confined to hydrogens: in acetyl fluoride the C=O bond comes out 44 mÅ short and C–F 36 mÅ long. With three numbers and fifteen parameters there is nothing in the data forcing any particular answer, and the fit is entitled to any of them.

7.4 Neither way of combining the sources is reliably better

The two hybrids get identical inputs and differ only in the merging rule, and which one wins is not predictable from anything known in advance. The *split* rule wins on fluoroethane at both data levels — including the case with the *most* unconstrained directions of any run here, fifteen of eighteen. The *joint* rule wins on vinyl fluoride and on sparse acetyl fluoride. Sorting the six cases by how underdetermined they are produces no pattern: split wins at deficits of 15, 3 and 1, joint at 12, 10 and 3.

This is a correction to an earlier result in this project. Run against isotopologue constants *derived* from the reference structures rather than measured, the same six cases produced a clean rule — joint when measurements are scarcer than parameters, split otherwise, holding in all six. That rule does not survive real data. Derived constants are mutually consistent by construction, which removes precisely the systematic zero-point offset that turns out to drive the behaviour, and the tidy pattern was an artefact of that.

7.5 The uncorrected zero-point offset is now the limiting factor

The results are no longer mainly limited by how the two information sources are merged. They are limited by fitting ground-state (r_0) constants with no vibration-rotation correction while scoring against a substitution (r_s) structure. That mismatch shows up as a one-signed bias of roughly half a percent in every measured B and C, and it is what the data-led fits spend their freedom chasing. It is also the one limitation here with a concrete fix: the correction is computable from a quantum force field, and this repository already implements the machinery. Doing so is the clear next step, and would change these numbers more than any change of objective.

7.6 Does the hybrid beat theory?

Since no rule reliably picks the objective in advance, the fair thing to report is the better of the two, while being explicit that the choice is made *after* seeing the answer and so overstates what could be achieved in practice. On a real problem both would be run — they cost seconds — but with no ground truth there would be nothing to choose between them.

Molecule	Data level	theory	best hybrid	which	change
Vinyl fluoride	parent only	19.0	19.5	joint	+0.5
Vinyl fluoride	all species	19.0	18.6	joint	-0.5
Acetyl fluoride	parent only	15.8	15.5	joint	-0.2
Acetyl fluoride	all species	15.8	8.8	split	-6.9
Fluoroethane	parent only	15.0	8.5	split	-6.5
Fluoroethane	all species	15.0	6.6	split	-8.4

RMS bond error in mÅ. Even choosing the better objective with hindsight, the hybrid beats theory in 5 of 6 cases — not all of them. On the C–F bond specifically it wins 6 of 6.

So the answer is a qualified no, and that is a change from what the same six cases showed on derived data. Bond accuracy overall: the hybrid is better in 5 of 6, and vinyl fluoride on parent data is a case where plain theory beats both hybrids and the experiment-only fit alike. The C–F bond is the exception that holds up — there some combination of data and theory is closer than theory alone every time, because theory's error there is a systematic bias and any real information about fluorine's position helps. But the C–F gain comes from the *split* objective and from the data-only fit; the joint prior stays within a few mÅ of theory on that bond, which is another way of saying it barely moves.

The honest summary is that on real measured constants, with no vibration-rotation correction applied, combining the two sources is not dependably better than using theory alone. That is a weaker claim than the derived-constant version of this experiment supported, and the difference between the two is the best argument in this report for insisting on measured data.

8. Limitations

What	Why it matters
No vibration-rotation correction is applied.	The measured constants are ground-state (r_0) values and are fitted as they stand, so the fit is pulled toward an r_0 structure while being scored against an r_s reference. The residuals show this plainly: B and C sit 0.3–0.7% off in the same direction in all three molecules, which is a real physical offset rather than scatter. It penalises the two data-using methods and leaves theory untouched. This repository implements the machinery to compute those corrections from a quantum force field; applying it is the obvious next step and would change these numbers.
Structure and constants come from different studies.	For vinyl fluoride the reference structure is a 1989 determination while the constants are the 1968 compilation of earlier work. They agree to 1.37%, inside the r_s -versus- r_0 range, but they are not one self-consistent experiment.

What	Why it matters
Some measured species had to be excluded.	Fluoroethane's seven multiply-deuterated species are omitted because the compilation's own configuration labels are ambiguous — it records that several configurations belong to the same isotopic species yet lists different constants for them — so which hydrogen each deuterium occupies cannot be established from the table. Including them on a guess would have been worse than leaving them out.
Hartree–Fock with a small basis is a weak level of theory.	A better method would shrink theory's C–F bias and narrow the gap. That would change the size of the effect, not its direction; the point here is the comparison between ways of using a fixed quantum surface, not the absolute accuracy of any of them.
Three different definitions of 'the structure' are in play.	The reference comes from isotopic substitution (r_s), the data are ground-state averages (r_o), and the quantum calculation produces the structure at the bottom of the energy well (r_e). These differ from one another by a few mÅ, so part of every error quoted here — for every method — is a definitional gap rather than a failure of the method.
Three molecules is a small sample.	The C–F bias is consistent across all three, and the objective-choice reversal reproduces a fourth system (fluorobenzene), but neither is established on a sample this size. They are indications worth acting on, not settled results.
One starting geometry per molecule.	Every method starts from the same displaced structure, which makes the comparison fair, but does not show how sensitive any of them is to where it starts.

9. Reproducing this

Both steps are in the repository and read no hidden state:

```
python scripts/monofluoro_benchmark.py
python scripts/build_monofluoro_report.py
```

Structures and constants live in `dev/monofluoro_references.py`, one entry per molecule with its citation. Every number in this report is read from the benchmark's JSON output rather than typed in, so the prose cannot drift from the run. The whole set takes about 5 minutes on one core.